

Sprint Scrum Planning

1. Project Overview

Project: HealthSphere - Digital Medical Platform

Duration: 20 weeks (5 months)

Team: 3 Master's students in Computer Engineering

Methodology: Scrum with 2-week sprints

Total Sprints: 10 sprints

2. User Stories & Story Points

Story Point Scale (Modified Fibonacci)

- **1 point:** Very small task (< 4 hours)
- **2 points:** Small task (4-8 hours)
- **3 points:** Medium task (8-16 hours)
- **5 points:** Large task (16-24 hours)
- **8 points:** Very large task (24-32 hours)

User Stories with Story Points

Authentication & User Profile Management

- **US1:** Patient registration and profile creation - **5 points**
- **US2:** Patient profile update with medical history - **3 points**
- **US12:** Doctor professional profile creation - **5 points**
- **US13:** Doctor profile update and document upload - **5 points**
- **US21:** Admin user account management - **3 points**

Doctor Search & Profile Management

- **US3:** Search doctors by criteria - **5 points**
- **US4:** Apply filters and sort results - **3 points**
- **US20:** View doctor reviews and ratings - **2 points**
- **US22:** Admin doctor credential verification - **3 points**

Appointment Booking & Management

- **US8:** Book appointment with doctor - **5 points**
- **US9:** View/reschedule/cancel appointments - **5 points**
- **US14:** Doctor availability and calendar management - **5 points**
- **US15:** Appointment request notifications - **3 points**

- **US16:** Doctor appointment management - **3 points**

Communication & Document Sharing

- **US5:** Initiate chat with doctor - **5 points**
- **US6:** View chat history - **2 points**
- **US7:** Upload medical documents - **3 points**
- **US17:** Doctor-patient communication - **3 points**
- **US18:** View and annotate patient documents - **3 points**
- **US19:** Generate digital prescriptions - **5 points**

Prescription & Medical Records

- **US10:** Receive and view digital prescriptions - **2 points**

Feedback & Reviews

- **US11:** Leave doctor reviews and ratings - **3 points**

Administrator Tools & Monitoring

- **US23:** Monitor system performance - **5 points**
- **US24:** Configure security policies - **3 points**

Total Story Points: 89 points

3. Sprint Breakdown

Sprint 0 - Project Setup (Weeks 1-2)

Sprint Goal: Establish development environment and project foundation

Backlog Items:

- Environment setup (Docker, databases, repositories) - **5 points**
- CI/CD pipeline configuration - **3 points**
- Database schema design and implementation - **5 points**
- API Gateway basic structure - **3 points**
- Project documentation setup - **2 points**

Sprint Velocity: 18 points

Key Deliverables: Working development environment, database schema, basic project structure

Sprint 1 - Core Authentication (Weeks 3-4)

Sprint Goal: Implement user authentication and basic profile management

Backlog Items:

- **US1:** Patient registration and profile creation - **5 points**
- **US12:** Doctor professional profile creation - **5 points**
- User authentication service (JWT) - **5 points**
- Basic frontend login/register pages - **3 points**

Sprint Velocity: 18 points

Key Deliverables: User registration, login functionality, basic profiles

Sprint 2 - Profile Management (Weeks 5-6)

Sprint Goal: Complete user profile functionality and admin tools

Backlog Items:

- **US2:** Patient profile update with medical history - **3 points**
- **US13:** Doctor profile update and document upload - **5 points**
- **US21:** Admin user account management - **3 points**
- Profile management frontend pages - **5 points**
- File upload functionality - **2 points**

Sprint Velocity: 18 points

Key Deliverables: Complete profile management, file uploads, admin panel basics

Sprint 3 - Doctor Search & Discovery (Weeks 7-8)

Sprint Goal: Implement doctor search and filtering capabilities

Backlog Items:

- **US3:** Search doctors by criteria - **5 points**
- **US4:** Apply filters and sort results - **3 points**
- **US20:** View doctor reviews and ratings - **2 points**
- **US22:** Admin doctor credential verification - **3 points**
- Doctor search frontend interface - **5 points**

Sprint Velocity: 18 points

Key Deliverables: Doctor search functionality, filtering, basic reviews

Sprint 4 - Appointment System Core (Weeks 9-10)

Sprint Goal: Build appointment booking system

Backlog Items:

- **US8:** Book appointment with doctor - **5 points**
- **US14:** Doctor availability and calendar management - **5 points**
- Appointment service backend - **5 points**
- Calendar integration frontend - **3 points**

Sprint Velocity: 18 points

Key Deliverables: Appointment booking, doctor calendar management

Sprint 5 - Appointment Management (Weeks 11-12)

Sprint Goal: Complete appointment lifecycle management

Backlog Items:

- **US9:** View/reschedule/cancel appointments - **5 points**
- **US15:** Appointment request notifications - **3 points**
- **US16:** Doctor appointment management - **3 points**
- Appointment management frontend - **5 points**
- Notification system - **2 points**

Sprint Velocity: 18 points

Key Deliverables: Complete appointment management, notifications

Sprint 6 - Communication System (Weeks 13-14)

Sprint Goal: Implement patient-doctor communication

Backlog Items:

- **US5:** Initiate chat with doctor - **5 points**
- **US6:** View chat history - **2 points**
- **US17:** Doctor-patient communication - **3 points**
- Chat backend service - **5 points**
- Real-time messaging frontend - **3 points**

Sprint Velocity: 18 points

Key Deliverables: Chat system, real-time communication

Sprint 7 - Document Management (Weeks 15-16)

Sprint Goal: Complete document sharing and prescription system

Backlog Items:

- **US7:** Upload medical documents - **3 points**
- **US18:** View and annotate patient documents - **3 points**
- **US19:** Generate digital prescriptions - **5 points**
- **US10:** Receive and view digital prescriptions - **2 points**
- Document management frontend - **5 points**

Sprint Velocity: 18 points

Key Deliverables: Document upload/sharing, digital prescriptions

Sprint 8 - Reviews & Admin Tools (Weeks 17-18)

Sprint Goal: Implement feedback system and admin monitoring

Backlog Items:

- **US11:** Leave doctor reviews and ratings - **3 points**
- **US23:** Monitor system performance - **5 points**
- **US24:** Configure security policies - **3 points**
- Review system frontend - **3 points**
- Admin dashboard - **4 points**

Sprint Velocity: 18 points

Key Deliverables: Review system, admin monitoring tools

Sprint 9 - Integration & Testing (Weeks 19-20)

Sprint Goal: System integration, testing, and bug fixes

Backlog Items:

- End-to-end testing - **5 points**
- Performance optimization - **3 points**
- Security testing and fixes - **3 points**
- UI/UX improvements - **3 points**
- Documentation completion - **2 points**
- Bug fixes and refactoring - **2 points**

Sprint Velocity: 18 points

Key Deliverables: Fully tested system, performance optimizations

6. Definition of Done (DoD)

Story Level DoD

- ☐ Feature implemented according to acceptance criteria
- ☐ Unit tests written and passing
- ☐ Code reviewed by at least one team member
- ☐ API documentation updated
- ☐ Frontend responsive design implemented
- ☐ Manual testing completed
- ☐ No critical bugs or security issues

Sprint Level DoD

- ☐ All story points completed or properly moved to backlog
- ☐ Sprint demo prepared and executed
- ☐ Retrospective completed with action items
- ☐ Code merged to main branch
- ☐ Deployment to staging environment successful

Release Level DoD

- ☐ All acceptance criteria met for release scope
- ☐ Performance testing completed
- ☐ Security audit passed
- ☐ User documentation completed
- ☐ Production deployment successful
- ☐ Stakeholder sign-off received
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- **Meeting notes:** Shared Google Docs
- **Progress tracking:** Jira/Azure DevOps/GitHub Projects